

Olek Osikowicz

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Education

- PhD** **University of Sheffield**, School of Computer Science Sheffield, UK
• Efficient and reliable simulation-based Autonomous Driving Systems evaluation
• *PhD Supervisors: Donghwan Shin & Phil McMinn* Sept 2023 – present
- BSc** **University of Sheffield**, Computer Science Sheffield, UK
• Graduated with **First-Class Honours** Sept 2020 – June 2023
• Dissertation: *Grounded In Reality Autonomous Driving Systems Testing*

Current Projects

Generative Modelling for Driving Scenarios

- Implemented a Variational Auto-encoder (VAE) in PyTorch to learn compact latent representations of driving scenarios
- Designed, trained and evaluated the model enabling efficient sampling of driving scenarios for safety-critical agent evaluation

Distributed Data Infrastructure

- Designed and deployed a multi-node computing cluster using Ray actors and AWS EC2 to parallelize software-in-the-loop evaluations
- Significantly improved the throughput of ML evaluation experiments in testing research group

Multi-Fidelity Scenario Optimization

- Designed Multi-Fidelity Bayesian Optimization algorithms for Autonomous Driving Systems (ADS) verification
- Reduced the computational cost of evaluating driving models by 16.8% compared to state-of-the-art baselines
- Maximized information gain on model behavior while minimizing simulation costs by varying simulator fidelity

Flaky Test Analysis for ADS

- Empirically discovered and analysed the causes of flaky tests in simulation-based ADS testing
- Reported and published mitigation guidelines to reduce the impact of flaky tests on ADS verification

Employment

- University of Sheffield**, Research Assistant in Simulation-Based Testing Sheffield, UK
• Collaborating with **KAIST (Korea Advanced Institute of Science and Technology)** June 2025 – present
• Project title: "Simulation-Based Testing for Mobility Cyber-Physical Systems of Systems"
• Developing automated Python tooling for large-scale ADS simulation and testing
- Dover Fueling Solutions**, Summer Intern Kraków, Poland
• Built and validated automated ETL data pipelines on Microsoft Azure June 2022 – Sept 2022
• Worked with SQL warehouses and Databricks for scalable data processing

Publications

- Multi-Fidelity Bayesian Optimization for Simulation-Based Autonomous Driving Systems Testing** June 2026

Olek Osikowicz, Phil McMinn, Wei Xing, Donghwan Shin

2026 *IEEE Intelligent Vehicles Symposium (IV 2026)*

CAWSR: Carla-AutoWare Scenario Runner

Jan 2026

David Gasinski, **Olek Osikowicz**, Gwilym Rutherford, Donghwan Shin

github.com/Intelligent-Testing-Lab/cawsr [↗](#) Under review at *Journal of Open Source Software (JOSS)*

Simulation Nondeterminism and Its Impact on Flaky Tests for Autonomous Driving Systems

Jan 2026

Olek Osikowicz, Phil McMinn, Donghwan Shin

Under review at *Automated Software Engineering (ASE) Journal*

Empirically Evaluating Flaky Tests for Autonomous Driving Systems in Simulated Environments

Apr 2025

Olek Osikowicz, Phil McMinn, Donghwan Shin

10.1109/FTW66604.2025.00009 [↗](#) 2025 *IEEE/ACM International Flaky Tests Workshop (FTW 2025)*

Skills

Programming: Python (advanced), C++, SQL, TypeScript, React Native, Svelte

Distributed computing: Multiprocessing, Ray, Docker, AWS (EC2/S3), GCP, SLURM, with Loki, Prometheus, Grafana

Machine Learning: PyTorch, Generative Models (VAEs), Bayesian Optimization, Reinforcement Learning

Languages: English (fluent), Polish (native), German (intermediate), Spanish (basic)

Teaching

Software Re-Engineering [↗](#)

Mar 2024 – present

Supporting undergraduate and master's students in re-engineering real-world Python projects.

Introduction to Algorithms and Data Structures [↗](#)

Feb 2023 – June 2023

Running tutorial sessions for first-year students, explaining the principles of modern algorithms and data structures.

Awards

Best teaching support award by School of Computer Science

July 2025

- Awarded by Head of School in recognition of excellence in teaching support as a Graduate Teaching Assistant in Software Re-engineering in the 2024/25 academic year.

University of Sheffield Billy Ibberson Bursary

Feb 2025

- Secured a competitive £1,771 grant awarded to support research that develops links between academic innovation and industrial practice.

Software Hut Prize

June 2023

- Awarded for the best software delivery to a real-world client; recognized for excellence in Agile development and effective team collaboration.

Service

Artifact Evaluation Committee Member for ESOP, FASE and FoSSaCS Artifact Evaluation

Jan 2026

- Reviewed and evaluated research artifacts for reproducibility.

Student Volunteer at International Conference on Software Engineering (ICSE 2025)

Apr 2025

- Assisted in organizing and managing conference sessions and events.